

Supplementary Material: Semantic Identity Compression

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
ACS1	adversary_forces_n_minus_1_queries AbstractClassSystem/Extended.lean	ACS5	model_completeness AbstractClassSystem/Extended.lean
ACS7	nominal_localization_constant_semantic AbstractClassSystem/Extended.lean	ACS8	shape_cannot_distinguish AbstractClassSystem/Core.lean
ACS9	shape_provenance_impossible AbstractClassSystem/Core.lean	ALT1	protocol_is_concession AbstractClassSystem/Core.lean
ALT2	protocol_not_alternative AbstractClassSystem/Core.lean	EMB1	semantic_non_embeddability AbstractClassSystem/Typing.lean
FRD1	ObserverModel.feasible_subsetMass_le_ optimalFeasibleMass Paper1IT/FiberRateDistortion.lean	FRD2	ObserverModel.optimalSubset_attains_ optimalFeasibleMass Paper1IT/FiberRateDistortion.lean
FXI1	fixed_axis_incompleteness axis_framework.lean	GPH13	Ssot.GraphEntropy.tagFeasible_iff_ maxFiberCard_le Paper1IT/GraphEntropy.lean
GPH16	Ssot.GraphEntropy.block_tagFeasible_iff_pow_ maxFiberCard_le Paper1IT/GraphEntropyAsymptotic.lean	GPH17	Ssot.GraphEntropy.maxFiberCard_blockObserve_ eq_pow Paper1IT/GraphEntropyAsymptotic.lean
GPH18	Ssot.GraphEntropy.blockTagRateBits_eq_mul_ oneShot Paper1IT/GraphEntropyAsymptotic.lean	GPH19	Ssot.GraphEntropy.blockTagRateBitsPerCoordinate_ eq_oneShot Paper1IT/GraphEntropyAsymptotic.lean
GPH20	Ssot.GraphEntropy.minBlockFeasibleAlphabet_eq_ pow Paper1IT/GraphEntropyAsymptotic.lean	GPH24	Ssot.GraphEntropy.feasibleAtAlphabetBase_iff Paper1IT/GraphEntropyAsymptotic.lean
GPH25	Ssot.GraphEntropy.maxFiberCard_le_one_of_ injective Paper1IT/GraphEntropy.lean	GPH27	Ssot.GraphEntropy.tagFeasible_one_iff_ injective Paper1IT/GraphEntropy.lean
GPH28	Ssot.GraphEntropy.maxFiberCard_mono_of_ factors_through Paper1IT/GraphEntropy.lean	GPH30	Ssot.GraphEntropy.uniformFiberErrorRate_ge_ one_sub_tag_ratio Paper1IT/GraphEntropy.lean
GPH32	Ssot.GraphEntropy.optimalFiberBitLength_ feasible Paper1IT/GraphEntropy.lean	GPH33	Ssot.GraphEntropy.optimalExpectedAdaptiveBitLength_ le Paper1IT/GraphEntropy.lean
GPH34	Ssot.GraphEntropy.conditionalEntropyGiven_le_ log2_mul_expectedAdaptiveBitLength Paper1IT/GraphEntropy.lean	GPH35	Ssot.GraphEntropy.exists_conditionalCodes_ expectedLength_le_entropy_bits_plus_one Paper1IT/GraphEntropy.lean
GPH37	Ssot.GraphEntropy.optimalFiberBitLength_mono_ of_factors_through Paper1IT/GraphEntropy.lean	GPH38	Ssot.GraphEntropy.taskRecoverable_iff_ maxTaskFiberCard_le_one Paper1IT/GraphEntropy.lean
GPH39	Ssot.GraphEntropy.maxTaskFiberCard_mono_of_ factors_through Paper1IT/GraphEntropy.lean	GPH40	Ssot.GraphEntropy.taskRecoverable_pair_iff Paper1IT/GraphEntropy.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
GPH41	Ssot.GraphEntropy.maxTaskFiberCard_pair_le_left Paper1IT/GraphEntropy.lean	GPH42	Ssot.GraphEntropy.observeFiber_prod_card Paper1IT/GraphEntropy.lean
GPH43	Ssot.GraphEntropy.maxFiberCard_prod Paper1IT/GraphEntropy.lean	GPH44	Ssot.GraphEntropy.maxFiberCard_eq_finiteSup Paper1IT/ComputabilityQuantization.lean
GPH45	Ssot.GraphEntropy.maxFiberCard_computable_by_fiber_enumeration Paper1IT/ComputabilityQuantization.lean	GPH48	Ssot.GraphEntropy.quantization_lower_bound_fin Paper1IT/ComputabilityQuantization.lean
GPH50	Ssot.GraphEntropy.precision_requirement Paper1IT/ComputabilityQuantization.lean	GPH52	Ssot.GraphEntropy.fiber_is_clique Paper1IT/GraphEntropy.lean
GPH53	Ssot.GraphEntropy.fiber_finset_is_clique Paper1IT/GraphEntropy.lean	GPH54	Ssot.QueryBitBridge.card_distinguished_set_le_two_pow_card Paper1IT/QueryBitBridge.lean
GPH55	Ssot.QueryBitBridge.clog_card_le_query_count Paper1IT/QueryBitBridge.lean	GPH56	Ssot.QueryBitBridge.fiber_card_le_two_pow_query_count Paper1IT/QueryBitBridge.lean
GPH57	Ssot.QueryBitBridge.maxFiberCard_clog_le_query_count Paper1IT/QueryBitBridge.lean	GPH60	Ssot.QueryBitBridge.distinguished_set_eq_full_transcript_capacity Paper1IT/QueryBitBridge.lean
GPH61	Ssot.QueryBitBridge.clog_query_floor_is_tight_of_full_capacity Paper1IT/QueryBitBridge.lean	GPH62	Ssot.QueryBitBridge.queryExpressivityGap_nonneg Paper1IT/QueryBitBridge.lean
GPH63	Ssot.QueryBitBridge.queryExpressivityGap_eq_zero_iff Paper1IT/QueryBitBridge.lean	GPH64	Ssot.QueryBitBridge.queryExpressivityGap_mono Paper1IT/QueryBitBridge.lean
GRC1	Ssot.Paper1IT.poissonCellCollisionProb_eq_one_sub_exp_mul_one_add Paper1IT/GrowthCollisions.lean	GRC3	Ssot.Paper1IT.poissonCellCollisionProb_pos_iff Paper1IT/GrowthCollisions.lean
GTB1	Ssot.Paper1IT.requiredTagBits_positive_iff_two_le Paper1IT/GrowthTagBudget.lean	GTB2	Ssot.Paper1IT.requiredTagBits_eq_zero_iff_le_one Paper1IT/GrowthTagBudget.lean
INF1	AbstractClassSystem.Scope.observer_factors AbstractClassSystem/Extended.lean	L1	matroid_basis_equicardinality axis_framework.lean
L4	l4_exchange_wrapper HandleAliases.lean	L5	l5_exchange_wrapper HandleAliases.lean
L6	nonorthogonal_complete_has_redundant_axis axis_framework.lean	L7	exists_semanticallyMinimal_subset axis_framework.lean
L8	exists_orthogonal_semanticallyMinimal_subset axis_framework.lean	LWDC1	LWDCConverse.collision_block_requires_bits lwd_converse.lean
LWDC3	LWDCConverse.maximal_barrier_requires_bits lwd_converse.lean	NOH1	admissibility_no_hidden_state discipline_migration.lean
NSL1	AbstractClassSystem.Neurosymbolic.neurosymbolic_zero_error_identity AbstractClassSystem/Neurosymbolic.lean	NSL2	AbstractClassSystem.Neurosymbolic.shape_alone_not_zero_error AbstractClassSystem/Neurosymbolic.lean
NSL3	AbstractClassSystem.Neurosymbolic.neurosymbolic_necessary_for_zero_error AbstractClassSystem/Neurosymbolic.lean	NSL4	AbstractClassSystem.Neurosymbolic.neurosymbolic_identity_recovery AbstractClassSystem/Neurosymbolic.lean
NSL5	AbstractClassSystem.Neurosymbolic.neurosymbolic_injective_implies_recoverable AbstractClassSystem/Neurosymbolic.lean	NSL6	AbstractClassSystem.Neurosymbolic.neurosymbolic_is_canonical_helper_view AbstractClassSystem/Neurosymbolic.lean
PD1	nominal_pareto_dominates_shape discipline_migration.lean	PRIV3	AbstractClassSystem.identity_disclosure_separation AbstractClassSystem/Extended.lean
PRV1	AbstractClassSystem.provenance_impossibility_universal AbstractClassSystem/Extended.lean	RDC1	ObserverModel.finiteRateDistortionConverse Paper1IT/FiniteRateDistortionConverse.lean
RDC2	ObserverModel.finiteConditionalRateDistortionConverse Paper1IT/FiniteRateDistortionConverse.lean	RDC3	ObserverModel.finiteObservationOnlyRateDistortionConverse Paper1IT/FiniteRateDistortionConverse.lean
RDC4	ObserverModel.finiteMinEntropyBudgetConverse Paper1IT/FiniteRateDistortionConverse.lean	RDC5	ObserverModel.uniformFiniteRateDistortionConverse Paper1IT/FiniteRateDistortionConverse.lean
RDC6	ObserverModel.finiteRateDistortionBound Paper1IT/FiniteRateDistortionBounds.lean	RDC7	ObserverModel.logBudgetLowerBoundFromError Paper1IT/FiniteRateDistortionBounds.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
RDC8	ObserverModel.observationOnlyRateDistortionConverse Paper1IT/RateDistortion.lean	RDC9	ObserverModel.observationOnlyMinEntropyBound Paper1IT/RateDistortion.lean
ROB1	AbstractClassSystem.OpenWorld.extend_to_force_barrier AbstractClassSystem/Undecidability.lean	ROB2	AbstractClassSystem.OpenWorld.barrierFreedom_not_extension_stable AbstractClassSystem/Undecidability.lean
UND1	AbstractClassSystem.OpenWorld.hasBarrier_not_computable AbstractClassSystem/Undecidability.lean	UND2	AbstractClassSystem.OpenWorld.barrierFree_not_computable AbstractClassSystem/Undecidability.lean
ZDT1	Ssot.Paper1IT.requiredTagBits_le_iff_le_pow Paper1IT/GrowthTagBudget.lean	ZEC1	Ssot.GraphEntropy.zeroErrorConditionalEntropySandwich Paper1IT/ZeroErrorConditionalEntropy.lean

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	Lean handle
Corollary IV.6: Asymptotic reading of the attribute-only witness cost	ACS1
Corollary V.11: Budget lower bound from distortion	RDC6, RDC7
Corollary III.17: Capacity collapse for representation-induced graphs	GPH53, GPH52
Definition III.8: Collision multiplicity	GPH28, GPH37
Corollary VI.7: Factorized product law	GPH43, GPH42
Corollary V.4: Fiberwise attainment	FRD2
Corollary II.10: Observable-state insufficiency	FXI1
Corollary III.14: Linear log-bit scaling	GPH19, GPH18, GPH24
Corollary VI.6: Helper-view monotonicity	GPH41
Corollary III.11: Maximal-Barrier Converse	LWDC3
Corollary VI.1: Pareto domination	INF1, PRV1, PD1, ALT1, ALT2, EMB1, ACS8, ACS9
Corollary II.8: Class identity is not representation-computable	PRV1
Corollary VI.4: Task-level coarsening monotonicity	GPH39
Corollary IV.8: Worst-fiber query lower bound	GPH56, GPH57
Corollary V.8: Zero-budget error floor	GPH30
Corollary III.24: Zero-error conditional-entropy sandwich	ZEC1
Corollary V.7: Zero-error threshold	ZDT1
Proposition IV.7: Binary query counting bound	GPH54, GPH55
Proposition III.16: Cluster-graph structure	GPH53, GPH52
Proposition IV.11: Existence of a minimal complete core	L7
Proposition IV.12: Canonical orthogonal core	L8
Definition IV.10: Derivability-aware axis presentation	L6
Definition III.21: Expected adaptive description length	GPH33, GPH32
Definition III.19: Representation-adaptive auxiliary description	GPH32
Definition III.10: Maximal-Barrier Regime	LWDC1
Theorem V.10: Fano-style finite converse	RDC1
Definition V.3: Fiberwise feasible subset	FRD1
Corollary III.23: Conditional-entropy lower bound	GPH35
Theorem III.13: Exact block feasibility law	GPH19, GPH18, GPH16, GPH24, GPH17, GPH20
Theorem III.12: Exact one-shot feasibility threshold	GPH13
Theorem VI.5: Helper-view sufficiency	GPH40
Definition III.2: Observation channel	ACS8
thm:info-barrier	INF1
Definition II.7: Coding model induced by a representation	INF1
Definition IV.5: Query family	ACS1
Theorem IV.13: Matroid of canonical distinguishing families	L8, L4, L5, L1
Theorem V.12: Observation-only min-entropy bound	RDC9
Theorem II.9: Observable-state sufficiency	ACS5
Definition VII.3: Finite realized world	ROB2, ROB1
Definition VII.6: Function-level barrier predicate	UND2, UND1
Theorem V.6: Semantic identity rate-distortion theorem	GPH30
Definition III.5: Auxiliary description	ACS7
Definition VI.3: Task-fiber ambiguity	GPH38

Auto summary: mapped 43/43 (full=43, derived=0, unmapped=0).